

## Quiz 6

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

**Problem 1.** (10 points) Let  $\mathbf{A} = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$  and  $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ .

(a) Reduce  $[\mathbf{A} \ \mathbf{b}]$  to  $[\mathbf{R} \ \mathbf{d}]$  (the reduced echelon form) to find the condition on  $b_1, b_2, b_3$  such that  $\mathbf{Ax} = \mathbf{b}$  is solvable.

$$\begin{aligned}
 [\mathbf{A} \ \mathbf{b}] &= \left[ \begin{array}{cccc|c} \textcircled{1} & 3 & 1 & 2 & b_1 \\ \textcircled{2} & 6 & 4 & 8 & b_2 \\ \textcircled{2} & -6 & 0 & 0 & b_3 \end{array} \right] \\
 &\rightarrow \left[ \begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & \textcircled{2} & 4 & b_2 - 2b_1 \\ 0 & 0 & \textcircled{2} & 4 & b_3 + 2b_1 \end{array} \right] \\
 &\rightarrow \left[ \begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right] \\
 &\rightarrow \left[ \begin{array}{cccc|c} 1 & 3 & 0 & 0 & -\frac{1}{2}b_2 + 2b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right] \\
 &\rightarrow \left[ \begin{array}{cccc|c} 1 & 3 & 0 & 0 & -\frac{1}{2}b_2 + 2b_1 \\ 0 & 0 & 1 & 2 & \frac{1}{2}b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 + 4b_1 \end{array} \right] = [\mathbf{R} \ \mathbf{d}]
 \end{aligned}$$

$$\mathbf{Ax} = \mathbf{b} \text{ is solvable} \Leftrightarrow b_3 - b_2 + 4b_1 = 0$$

(b) Find the complete solution to  $Ax = b$  with  $b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ .

When  $b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$

$$[A \ b] = \left[ \begin{array}{cccc|c} 1 & 3 & 1 & 2 & 1 \\ 2 & 6 & 4 & 8 & 3 \\ -2 & -6 & 0 & 0 & -1 \end{array} \right]$$

pivot variables:  $x_1, x_3$   
 $\Rightarrow$  free variables:  $x_2, x_4$

$$\rightarrow \left[ \begin{array}{cccc|c} \textcircled{1} & 3 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \textcircled{1} & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] = [R \ d]$$

i) Solve for particular solution.

$$RX_p = d \Rightarrow \begin{cases} x_1 + 3x_2 = \frac{1}{2} \\ x_3 + 2x_4 = \frac{1}{2} \end{cases}$$

let  $x_2 = x_4 = 0$ , we get  $x_p = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}$

ii) Solve for special solution

$$RX_h = 0 \Rightarrow \begin{cases} x_1 + 3x_2 = 0 \\ x_3 + 2x_4 = 0 \end{cases}$$

$$x_2 = 1, x_4 = 0 \Rightarrow S_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$x_2 = 0, x_4 = 1 \Rightarrow S_2 = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Thus

$$x = \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} + C_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - 3C_1 \\ C_1 \\ \frac{1}{2} - 2C_2 \\ C_2 \end{bmatrix}$$